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Multispecies architecture in post-pandemic urban nature - Three typologies of connectedness

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ABSTRACT: The Covid19 pandemic has sparked a renewed focus on multispecies connectedness. Current architecture and design research tends to prompt design for multispecies cohabitation, yet we must also study the multifaceted issues already arising from connections between humans and other species. We therefore explore how multispecies connectedness is created and maintained by way of the built environment through field studies in Copenhagen's post-pandemic urban nature. We analyse six examples of multispecies architecture and launch three typologies: (pre)cautionary exclusion, pragmatic co-existence and staged connectedness.

1 INTRODUCTION

new focus on urban nature. During pandemic lockdowns, dolphins and other wildlife turned up in the human-less cities and citizens noticed how birdsong could suddenly be heard. Urban nature and biodiversity became more prevalent foci in discourse around policy, planning, and climate activism. In many places, especially in densely populated areas of the global North, COVID also changed people's access to and use of urban nature which, except for periods of severe stay-at-home orders, increased intensely. Demand rose for green features near the home and surveys show continued increase in people's use of green spaces continued post-pandemic. A considerable body of research has documented relationships between urban nature and human wellbeing and mental health during lockdown. Due to COVIDs zoonotic origin, the pandemic was also an occasion to reconsider multispecies entanglement of humans, vira and animals including how we in our cities and built environments relate to other species. Hence, as Bruno Latour has argued, the pandemic is not to be seen as a crisis, but rather as a reminder of a general condition: We share the Earth with other species including vira and must develop a new modus vivendi (Latour, 2021).

In this paper we explore how multispecies connectedness is created and maintained by way of the built environment based on field studies in Copenhagen's post-pandemic urban nature. The research undertaken is part of the RESPOND project focusing on the long-term consequences of the Covid-19 pandemic on everyday life and the built environment in Denmark. We have elsewhere discussed the battle for green space in the pandemic and post-pandemic dense city and concluded that interdependency between human and other species must be a key focus of future research(Nordberg & Stender, 2025). In this paper, we therefore emphasize not just human needs for nature, but rather how multispecies relations are mediated and maintained by way of the built environment. We are inspired by post-human and new materialist scholars that have promoted the concept of care in architecture (MOULD, 2022; Tronto, 2019). This entails thinking of buildings in relationships with ongoing environments, people, flora and fauna(Tronto, 2019:28). Architecture – here understood broadly as built environments including urban nature – can often determine how people, animals and plants share space – who and what is excluded, welcomed, exploited, and cared for (MOULD, 2022:8). This perspective may be an outset for promoting a 'kind architecture' heightening the perception of human-nonhuman entanglements in built environments(Saeidi, Anderson, & Davidová, 2023). Yet we seek to contribute by problematizing a narrow focus on kindness through exploring the frictions and unintended aspects of connectedness in the built environment. We recognize architecture as a political

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endeavour where care, power and conflict are interwoven by analysing "a world multiple" in which both humans and non-humans participate in the world making (Tsing, 2019:238).

The paper starts with a brief outline of our position within existing research on multispecies architecture and (post)pandemic urban nature. Then follows an account of the methods applied. After this comes our analysis of 6 examples of multispecies architecture in post-pandemic Copenhagen. Finally we present a concluding discussion, where we launch three typologies of multispecies architecture to build on in future research.

2 MULTISPECIES ARCHITECTURE AND POST-PANDEMIC URBAN NATURE

The pandemic and post-pandemic situation fostered an array of new discussions of both benefits and dangers related to multispecies cohabitation in urban nature. When focusing on how the pandemic has affected relationships between humans and other species, who (also) have the city and urban nature as their habitat, several streams of inquiry are observed. Biological, zoological and ecological research treat themes such as ecological effects of COVID (Gaiser et al., 2022), wildlife reactions to human-less cities during lockdown (Gibbs, 2022) and bird reactions to humans wearing face masks (Mikula et al., 2021). Within the humanities studies show how lockdowns affected human reactions to urban wildlife (Murray et al., 2022) and how COVIDs zoonotic origin affected the perception of urban companion animals (Yin, Gao, Zhu, & Li, 2020).

COVID has also been examined as a more-than-human situation, a multispecies entanglement of humans, vira and animals (Gibbs, 2022; Lunstrum et al., 2021; Mcintyre-Mills, 2021; Royle, 2020) or an even bigger network of actors including a multitude of animal, plant and fungi species, microorganisms, ecosystems, air, technologies, institutions, societal structures, media, politics etc. (Faas et al., 2020; Iared & Hofstatter, 2022; Meloni & Maller, 2024; Searle, Turnbull, & Lorimer, 2021). Some of these studies add emphasis to how COVID affected eg. the city as a shared habitat for human and non-human animals (Mcintyre-Mills, 2021), multispecies justice (Celermajer & Mckibbin, 2023), or how the pandemic caused an "anthropause" in an otherwise anthropocentric world (Searle, Turnbull, & Lorimer, 2021).

This variety of studies related to COVID and multispecies cohabitation points towards a need to pay even closer attention to the relationships mediated by multispecies architectures, both as an answer to growing humans needs to connect with other species and to prepare ourselves and our urban nature for what comes next in the modus vivendi of polycrises. Current literature tends to focus on benefits and promotion of multispecies connectedness and cohabitation and especially when it comes to design, promotes speculative or utopian characteristics. Instead, we seek to analyse the variety and conflicts of multispecies connectedness through very concrete material architectures by combining the analysis of physical multispecies architecture, with the critical view on more than human entanglements. Thereby being attentive to the asymmetries of power inherent in anthropogenic urban nature and affecting the possibility for caring architecture.

3 METHODS AND CASES

As Tsing suggests, we may study non-human practices in similar ways as we study humans: by observation and attention to forms, gatherings, and transformations(Tsing, 2019:238) . Talking to scientists but also non-scientists with different types of knowledge can give valuable insight into various non-human practices – for Tsing mushroom pickers were key, for us the maintenance workers were essential. We have in each of the three cases therefore conducted recurrent field studies with observations, registrations, and interviews, both with human users, staff and maintenance workers. Furthermore, we have visited each of the three cases with a biologist researcher specialized in urban nature helping us identify other present species, their living conditions and practices. The three cases were chosen because they are all examples of urban nature spaces in dense neighbourhoods of Copenhagen, yet they are also quite different in terms of how connectedness between human and no-human actors unfold. Common for the three areas was that they experienced increased human use and conflicts between users (human and non-human) during times of COVID restrictions. Each area will be described further in the analysis.

During the process of gathering and analysing data our attention was turned to particular points of interest: sections of the built environment, where multispecies relations were particularly at play. In the following analysis we therefore focus on these 6 points – two for each case. For each of them we have created a diagram. As Tsing also notes, diagrammatic expositions can be helpful in showing the coordinations involved in keeping multispecies assemblages in place (Tsing, 2019:242). Another useful aspect of diagrams, also noted by Tsing(2019:242), is allowing several times or timescales to be present in the same diagram. The diagrams thus incorporate both our "in the moment" observations, the practical and historical descriptions by operating staff and the biologist view on the cyclical timescales of animals, plants and ecosystems.

3.1 *The Lakes*

Both Copenhageners and tourists enjoy strolling on the pedestrian promenades surrounding the lakes, and during COVID they were so popular that unidirectional traffic had to be introduced. As features of urban nature, there is a difference of typology between the southernmost Lake, St. Jørgens, and the two others since St. Jørgens Lake has more natural lake shores with vegetation while Peblinge and Sortedam Lake have manmade straight kerbstone edges. This distinction also exists in the operation of the Lakes, where St. Jørgens Lake is allowed to look "wilder" and is not extensively maintained, while the other two Lakes are more architectural and park-like and therefore more strictly kept. COVID challenged both the operation and recreational use of the Lakes, pedestrians strayed from the paths to avoid each other causing extreme tear on the planting and even resulting in claims for damages, when people stumbled over roots on their irregular "desire-paths".

3.1.1 *The swan fence*

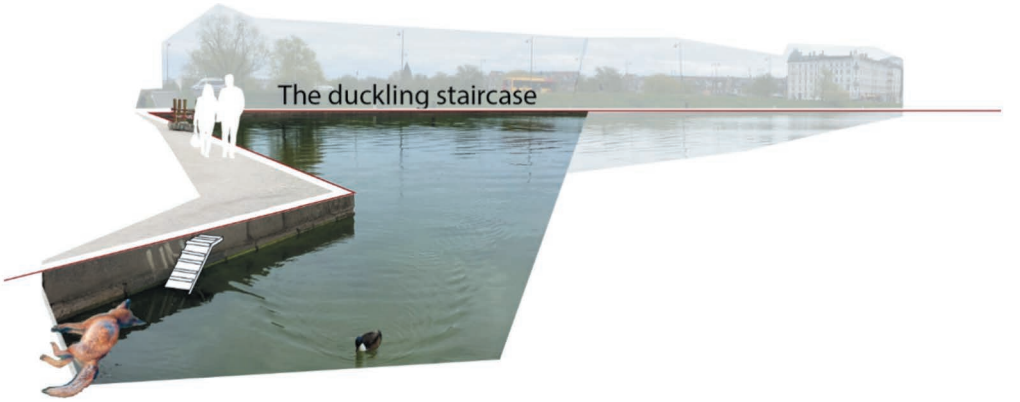
Along the side of St Jørgens Lake, a makeshift fence made of iron poles and elastic rope has been erected. It is situated on the verge where the path for bikes and pedestrians meets the grass and rush vegetation of the lakeshore. On the other side of the fence a swan is brooding in a large nest only sparingly hidden behind the rush. The park steward made it clear that the swans have a special status to the users of the Lakes, extensively photographed, fed with stale bread and advocated for, and somewhat caricatured by the swan shaped pedal boats on Peblinge Lake. It is not defined exactly, who the fence is for – is it to keep pedestrians from disturbing the brooding swan, when taking photos or to avoid altercations between swans known to be aggressive and innocent passers-by? One thing the fence did not manage to keep out was a rat running across the path to the nest, maybe to eat crumbs left for the swan or other leftovers in the nest.



3.1.2 *The duckling staircase*

The steep vertical sides of Peblinge and Sortedam Lakes means a very abrupt transition between the architectural promenade and the water as animal habitat. This transition causes problems for the birdlife, especially the young of ducks, swans, coots, and grebes who cannot scale the steep edge and get to shore until they learn to fly. Several places along Peblinge and Sortedam Lakes, duckling staircases have been installed, to mediate the hard edge between promenade as

architecture and water as habitat. The stringency of the lake architecture is an aesthetic choice for the benefit of humans, but the (naturally) un-aesthetic appears as an unintended consequence – the park steward told of complaints from human users when a drowned fox was found in the lake. Following its natural urge to hunt waterfowl it had jumped in and died from not being able to get out again. Human users often prefer urban nature to be devout of the un-pleasant aspects of nature t – death, decay and the stench that follows are considered harmful especially in pandemic times with a raised hygiene-focus. The duckling staircases both offer aid for young waterfowl and for the city dwelling humans, who prefer not to see the death so ubiquitous in nature.



3.2 *The City of the Old*

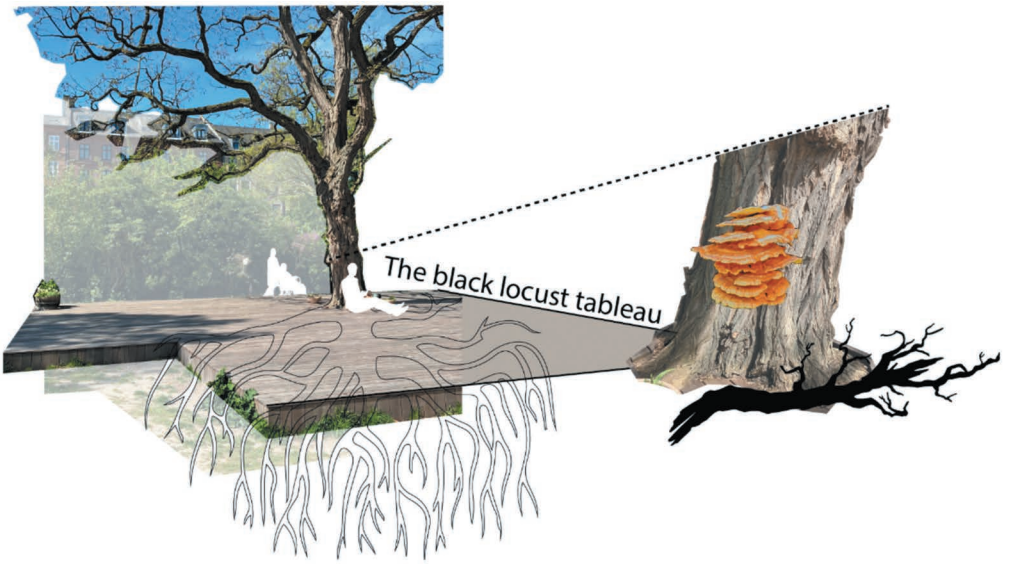
The City of the Old contains several retirement homes and assisted living facilities, more than five daycare institutions for children, refugee housing, medical facilities, a church, a recycling centre and a farm-playground with goats, chickens, rabbits and other critters and community gardens, all located in a park like area with old growth trees and lawns. Besides the inhabitants, it is widely used by the neighbours: dog walkers, picnickers, exercisers, partying youths, drug dealers and the homeless. During COVID the neighbours were even more active in the area, resulting in both conflicts such as nursing home staff being fearful of the growing amounts of drug dealers and the operation staff experiencing massive growth in the amount of trash.

3.2.1 *The black locust tableau*

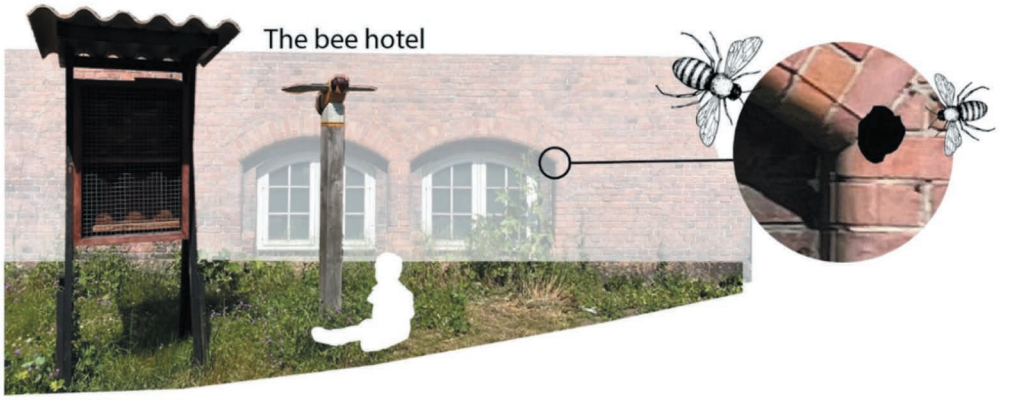
The playground area has only recently re-opened after a renovation necessitated by soil pollution. The top layer of soil has been removed and exchanged with new soil except from the area surrounding an old black locust tree, where a wooden tableau has been established. The operating staff explained that the roots of the black locust could not tolerate the disturbance caused by an exchange of soil, so the tableau had been established instead. The plateau attracts playground visitors, offering seating along the edge and a level surface for picnicking. The biologist noted that the black locust was not a native plant and upon closer inspection, saw that the tree was host to the wood decaying fungi "chicken of the woods" (Laetiporus sulphureus). While the plateau had been built to protect the tree from disturbance and the human users from polluted soil, the tree in itself is now a "risk-tree" since the fungal decay can cause branches or the entire tree to fall. The status as risk-tree means that the operating staff will soon have to cut down the tree for the safety of humans, even though a decaying tree offers excellent habitat for other species, according to the biologist especially the endangered bats, which could roost in the cavity caused by this type of fungal decay in a black locust.

3.2.2 *The bee hotel*

In a corner of the City of the Old we come across a strange little shed-like structure and as a symbolic explainer a large wooden bee is placed on an adjacent post. The bee hotel as the shed is called, is intended as nests for solitary bees. It consists of a roofed cabinet on legs with a chicken wire door and a collection of wooden tubes inside. According to the biologist, both the wooden bee and the bee



hotel mainly had a symbolic character, since the solitary bees in the area were not in need of nesting places, any cavity in buildings or soil would be just fine, but much more in need of an entire ecosystem, especially a variety of native plants to harvest pollen from. Social media also showed users who were worried that attracting more bees to a playground would be dangerous for children. The operating staff said that there had previously been urban beekeeping hives in the area, but that they had been moved because of the many elderly people with dementia living in the area. The bees are on one hand a strong symbol of biodiversity preservation used in the nature education taking place around the playground and on the other hand animals that are considered dangerous to vulnerable humans such as infants and the elderly.



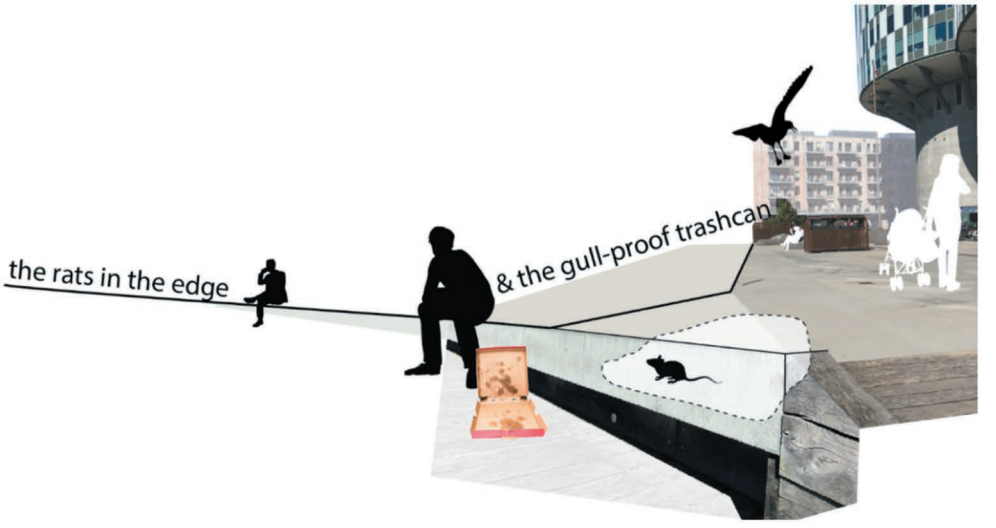
3.3 *The Aarhusgade neighbourhood*

In the former industrial harbour Nordhavn a new urban area was developed between 2013 and 2022 called the Aarhusgade neighbourhood, the first phase of a large redevelopment and land reclamation. Designed to house 2000 inhabitants and workplaces for another 7000 the area is characterised by high building density and the closeness to water, with a harbour bath, quay promenades and a coastal theme in the design and plant choice of the sparse plots of urban nature. During COVID and beyond, especially the harbour bath has drawn large amounts of visitors to the area to swim, sunbathe and party alongside the diverse wildlife that frequent areas where land meets sea – harbour porpoises, food fish such as garfish or mackerel, waterfowl and rats. The operations manager even described how plant- and wildlife such as fry and eelgrass was planted out, to achieve the

nature desired by the human users. Triggered by the ”vira-phobia” caused by COVID and the massive influx of visitors, the locals became very preoccupied with cleanliness in the area, complaining about trash, droppings and pests – rats and seagulls – and demanding a clean urban nature.

3.3.1 *The rats in the edge and the gull-proof trash can*

öteborg Square is the central open space in the area, placed next to the harbour bath and surrounding a converted silo acting as an architectural landmark. The square is covered in concrete slabs but towards the water and the harbour bath it terraces into a wooden construction functioning as seating accommodation for the very large number of visitors flocking to the area, when the weather is nice. For the human inhabitants this influx of visitors leads to a number of nuisances that can be described as the smelly side of human nature, urination, defecation, used hygiene products and other perishable forms of litter. Human littering also provides excellent living conditions for unwanted urban wildlife. When built, the quay edge was constructed in a way that allowed for rats to enter and build nests that undermined the concrete slabs of the square and the operations manager recalls how, when they opened it up, the nests were predominantly built from takeaway food wrappings. The reconstructed quay edge is now rat-proof and similarly a protective structure had to be built to cover the many waste containers and thereby avoid that human litterbugs were joined by seagulls redispersing litter in the area in their search for food.

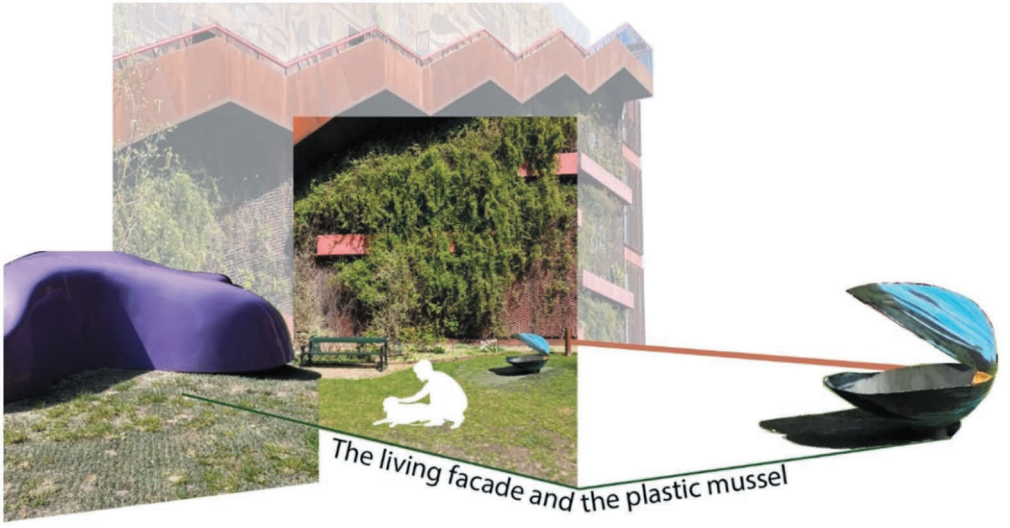


3.3.2 *The living façade and the plastic mussel*

Away from the quays and wedged in between apartment buildings and a parking structure you find the playground at St. Petersburg Square. The façade of the parking structure facing the playground is constructed as a “vertical garden” with twining plants such as ivy and clematis growing from planter boxes on a perforated Corten steel cladding. Even though the foliage makes the façade look “green”, it does not - according to the biologist - support any wildlife in particular. Birds might perch in it, which we saw and especially heard on our visit, but it has few qualities as habitat or source of nourishment as the plants are not native to the surrounding ecosystem. As a resource for the human inhabitants, the square is also difficult to inhabit, it is described by a resident as “the slimy playground” since the flower beds and lawns seem to have issues with draining excessive rainwater. As witnessed by the amount of signs reminding people not to let their dogs defecate, the playground is one of very few places in the area, where children and dogs can feel the grass under their feet. This leads to such intensive use that, in combination with poor growth conditions, the lawns are patchy, withered and in places supplemented by artificial turf. The playground elements made to look like gigantic marine life – a giant purple starfish and a plexiglass mussel you can fit a child in – adds to the feeling that in this park urban nature is quite metaphorical. At the same time the urban nature elements demand intensive maintenance to survive. A man is working full time solely with the

vertical garden, not just keeping it alive, but also cutting it from the inside, so it does not overgrow the parking lots. The façade facing the playground is also the only one of the structure's four façades, where it has been possible to achieve favourable enough growth conditions to keep foliage alive.

4 CONCLUDING DISCUSSION



Our three sites are places where biodiversity is cultivated in ways that could be called slow disturbance to use Anna Tsings term for "anthropogenic [human-made] ecosystems in which many other species can live"(Tsing, 2012). Across the six cases we have observed three different typologies of connectedness between species (humans and other) becoming apparent in specific instances of multispecies architectures. The typologies are not mutually exclusive or fixed but allows us to discuss, how architectures foster different types of connectedness between species.

The first typology of connectedness, we identify, can be describes as (pre)cautionary exclusion. When the edge of Göteborg plaza towards the water was redone to stop rats from building nests out of discarded food wrappers and undermining the concrete slabs of the plaza or when gull-proof cages are built to cover trash cans, human needs call for architectures mediating cautionary or sometimes precautionary exclusion. For humans, some elements of nature and some species are considered hazardous or unhygienic enough to warden exclusion, even in parts of the built environment of urban nature, where multispecies cohabitation is otherwise sought for and maintained. Exclusion is of course a negative connectedness, but it is an important part of the relationship that what causes the human need for exclusion is that these other species, rats and gulls thrive in an environment, where waste left by humans is abundant.

The second typology could then be named pragmatic co-existence. Architectures mediating pragmatic coexistence are often seen as lines of demarcation or barriers, not unlike (pre)cautionary exclusion but to a higher degree offering mutual benefit or "giving space" to other species. We found architecture fostering pragmatic coexistence in the way the swan fence could either protect swan from man or man from swan. Similarly, in the City of the Old, where humans needed to be protected from toxic soil and the black locust needed to be protected from having its roots disturbed and the plateau was constructed to see to both needs.

These typologies are not temporally fixed – the tableau loses its purpose, when the Black Locust is attacked by fungi and becomes a hazard-tree warranting an ultimate "precautionary exclusion" despite that the tree-fungi relationship could create valuable habitat for other species – insects, birds and in particular bats. The duckling staircase helps ducklings and foxes exit the lake, even though humans have built a lake edge that excludes such passage, but one could argue that the primary function is to help humans avoid the discomfort of seeing a drowned fox or duckling.

The third type is what we call staged connectedness. This would be examples like the bee hotel, the vertical garden and plastic mussels of St. Petersburg Square. Such architectures showcase a human relationship of engagement to nature, flora and fauna, while not really caring for or establishing real habitats for other species. The areas of urban nature we have examined all have very little potential to radically affect the global biodiversity crisis. Both the biologist, nature playground staff and the park steward point out that the positive effect of urban nature is connecting human city dwellers with other species educationally and emotionally to foster a general appreciation and understanding of nature more than actual biodiversity and ecosystem preservation. Therefore the staged connections are not without care, but are balancing a line of communicating nature to foster connection and "greenwashing" human-centric architecture. If we want to care for other species through architecture, typologies of connectedness help us understand the relationships we are already mediating through our architectures and the conflicts and synergies arising. Only through understanding of the asymmetrical needs of humans and other species can we "turn our [architectural] power towards caring for our broken world" as Tronto argues. While recent scholarship promotes 'kind architecture' heightening the perception of human-nonhuman entanglements, the three typologies may encourage future research and practice to pay attention also to unintended frictions and unequal power relations in built multispecies connectedness.

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